

For your Intel Mac mini 2012, do it in this order.

1. Install Homebrew

Open Terminal:

```
/bin/bash -c "$(curl -fsSL  
https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

Verify:

```
brew --version
```

2. Install Python + Tesseract

```
brew install python@3.12
```

```
brew install tesseract
```

Verify:

```
python3 --version
```

```
tesseract --version
```

3. Copy the SAM project to the Mac

For example:

```
~/sam-ticket-server
```

Copy:

- `app.py`
- `ticket_detector.py`
- `detect_tickets.py`
- `requirements.txt`
- `models/`
- everything needed

Especially:

```
models/sam_vit_b_01ec64.pth
```

4. Create virtual environment

Go to project folder:

```
cd ~/sam-ticket-server
```

Create venv:

```
python3 -m venv .venv
```

Activate:

```
source .venv/bin/activate
```

5. Upgrade pip

```
pip install --upgrade pip setuptools wheel
```

6. Install PyTorch CPU

For Intel Mac:

```
pip install torch torchvision
```

7. Install project requirements

```
pip install -r requirements.txt
```

8. Set environment variables

```
export SAM_API_KEY="change-me"  
export SAM_DEVICE="cpu"
```

9. Start the server

```
python -m uvicorn app:app --host 0.0.0.0 --port 8000
```

You should see:

```
Application startup complete  
Uvicorn running on http://0.0.0.0:8000
```

10. Test locally on Mac

Open another terminal:

```
curl http://127.0.0.1:8000/health
```

Then test image upload:

```
curl -X POST "http://127.0.0.1:8000/detect-tickets" \  
-H "x-api-key: change-me" \  
-F "image=@/path/to/image.jpeg"
```

11. Make it reachable publicly

Best option:

- Cloudflare Tunnel

Install:

```
brew install cloudflared
```

Login:

```
cloudflared tunnel login
```

Temporary public tunnel:

```
cloudflared tunnel --url http://localhost:8000
```

It will give you a public HTTPS URL like:

```
https://random-name.trycloudflare.com
```

Then your Azure/.NET backend can call:

```
POST https://random-name.trycloudflare.com/detect-tickets
```